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QUARTERLY EXAMINATION - 2026

12 Std

CHEMISTRY

Reg No.

Time : 3.00 Hours

MARKS: 70

PART - I

I Choose the correct answer. **15x1=15**

1. Which of the following is used for concentrating ore in metallurgy?
   a) Leaching b) Roasting c) Froth floatation d) Both (a) and (c)
2. The following element is not refined by zone refining porcess is
   a) Ge b) Sr c) Si d) Ga
3. The basic structural unit of silicates is
   a) $(SiO_3)^{2-}$ b) $(SiO_4)^{2-}$ c) $(SiO)^-$ d) $(SiO_4)^{4-}$
4. The strongest acid among all is .....
   a) HI b) HBr c) HF d) HCl
5. Which one of the following is amphoteric
   a) CrO_3 b) Cr_2O_3 c) CrO d) None of these
6. Which one of the following ions has the same number of unpaired electrons as present in V^{3+}?
   a) Ti^{3+} b) Fe^{3+} c) Ni^{2+} d) Cr^{3+}
7. The ratio close packed atoms to tetrahedral hole in cubic packing is
   a) 1:1 b) 2:1 c) 1:2 d) 1:4
8. The crystal with a metal deficiency defect is
   a) FeO b) NaCl c) ZnO d) KCl
9. The half life period of first order reaction is
   a) $t_{½} = \frac{2.303}{k} \log 2$ b) $t_{½} = \frac{2.303}{k} \times 0.3010$ c) $t_{½} = \frac{0.693}{k}$ d) All of these
10. Which of the following fluro compounds is most likely to behave as a Lewis base?
    a) BF_3 b) PF_3 c) CF_4 d) SiF_4
11. The aqueous solutions of sodium formate, anilinium chloride, and potassium cyanide are respectively
    a) acidic, acidic, basic b) basic, acidic, acidic c) basic, neutral, basic d) None of these
12. Assertion : Phenol is more acidic than ethanol
    Reason : Phenoxide ion is resonance stabilized
    a) Both assertion and reason are true and reason is correct explanation of assertion
    b) Both assertion and reason are true and reason is not the correct explanation of assertion
    c) Assertion is true but reason is false d) Both assertion and reason are false
13. Cold alkaline solution of $KMnO_4$ is
    a) Fenton's reagent b) Tollen's reagent c) Baeyer's reagent d) Lucas reagent
14. Which one of the following reaction is an example of disproportionation reaction.
    a) Aldol condensation b) Cannizaro reaction c) Benzoin condensation d) None of these
15. The formation of cyanohydrin from acetone is an example of
    a) Nucleophilic substitution b) Electrophilic substitution
    c) Electrophilic addition d) Nucleophilic addition

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II Answer any Six of the following. Question number 24 is compulsory. **6x2=12**

16. Which type of ore can be concentrated by froth floatation method? Give two examples.
17. How will you convert boric acid to boron nitride?
18. What type of hybridisation occur in
    (a) BrF_5 (b) BrF_3
19. Which is more stable Fe^{2+} or Fe^{3+} ? Why?
20. Define unit cell.
21. Identify the order for the following reactions.
    (i) Rusting of Iron (ii) Radioactive disintegration of $_{92}U^{238}$
22. Give the uses of diethyl ether.
23. Write a note on Rosenmund reduction.
24. Write the expression for the solubility product of (a) $BaSO_4$ (b) Ag_2CrO_4

PART - III

III Answer any Six of the following. Question number 33 is compulsory. **6x3=18**

25. What are the differences between minerals and ores.
26. Write the uses of Borax.
27. What are Interhalogen compounds? Give two examples.
28. Write Hume - Rothery rule for formation of alloys.
29. Write a note on Frenkel defect.
30. Derive integrated rate law for zero order reaction. $A \rightarrow$ Product
31. What is buffer solution? Mention the two types of buffer solution with example.
32. What is Urotropine? How is it prepared?
33. How are glycerose prepared from glycerol?

PART - IV

IV Answer all the questions. **5x5=25**

34. (a) (i) Give the basic requirement for vapour phase refining. (2)
    (ii) Write the applications of zinc. **[OR]** (3)
    (b) Explain the structure of Diborane. (5)
35. (a) (i) Write the reason for the anomalous behaviour of Nitrogen. (2)
    (ii) Explain the Deacons's process for manufacture of chlorine. **[OR]** (3)
    (b) Compare Lanthanoids and actinoids. (5)
36. (a) What are the difference between amorphous and crystalline solids? **[OR]** (5)
    (b) (i) Explain Pseudo first order reaction with an example. (3)
    (ii) Define order of a reaction. (2)
37. (a) Derive an expression for Ostwald dilution law. **[OR]** (5)
    (b) Explain Sayzteff's rule with example. (5)
38. (a) (i) Distinguish between primary, secondary and tertiary alcohols using Lucas reagent. (3)
    (ii) Write any two test for carboxylic acid group. **[OR]** (2)
    (b) Write the mechanism of Aldol condensation reaction. (5)

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